

BLUETOOTH CAR

Thesis Submitted in
Partial fulfillment for the award of

Bachelor of Science Computer Science & Information Technology

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CERTIFICATE

It is certified that the project titled **Smart Clean Admin** has been carried in group are Bonafide students of BSC(CS/IT) VI semester, Department of Computer Science, Silver Oak University, Ahmedabad, Gujarat. The project is a record of the work accomplished during the sixth semester of BSC(CS/IT) as a partial fulfillment of the requirement for the award of degree in Bachelor of science in computer science and information technology (BSC CS-IT), under my guidance.

STUDENTS

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DECLARATION

We hereby declare that project entitled “ **RC CAR** ” in partial fulfilment for the degree of **Bachelor of Computer Science** to **Silver Oak University**, Ahmedabad, is bonafide record of the project work carried out at **SILVER OAK COLLEGE OF COMPUTER APPLICATION** under the supervision of **Mr.Ankit Patel** and that no part of any of these reports has been directly copied from any student’s report or taken from any other sources, without providing due reference.

Name of students	Sign of student
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MAYANK SONAGARA	
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AKSHAR PATEL	
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BHARGAV PATEL	
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UTTAM OGANIYA	
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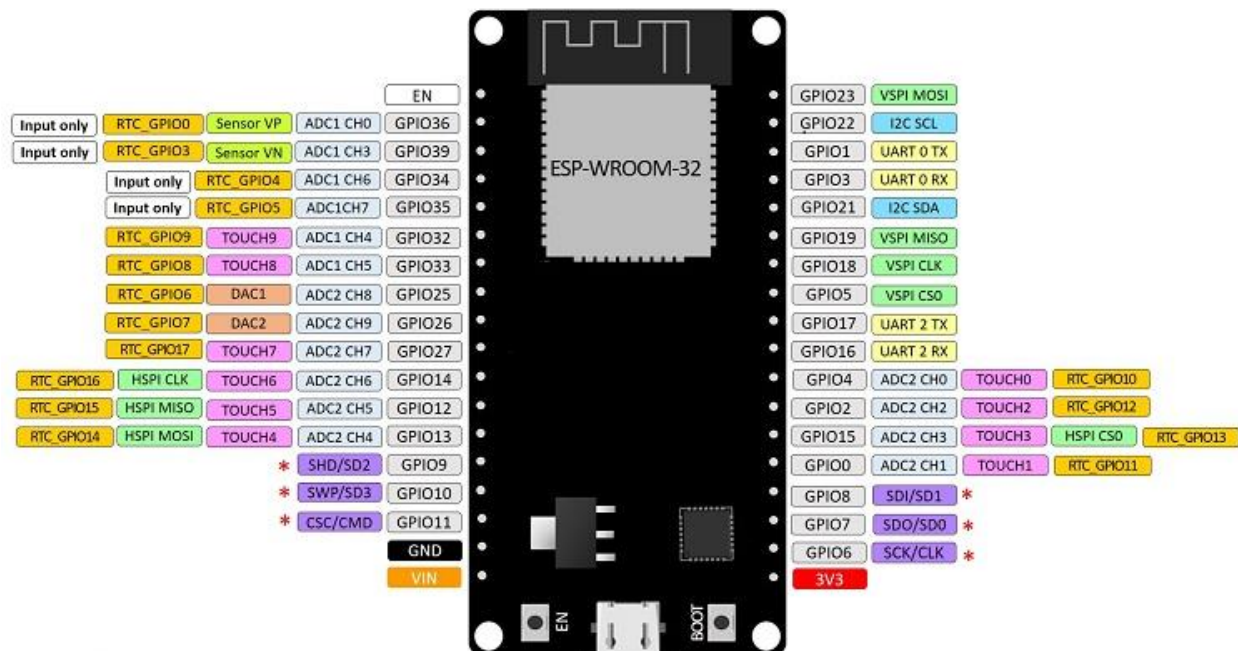
BHAVYESH PANCHAL	
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COMPONENTS

- 1. ESP32 Microcontroller**
- 2. L298N Motor Driver**
- 3. DC Motors**
- 4. Bluetooth Module (HC-05)**
- 5. Power Supply (Battery)**
- 6. Jumper Wires**
- 7. Breadboard**
- 8. Car Chassis**
- 9. Wheels and Motor Assembly**
- 10. Car Body**

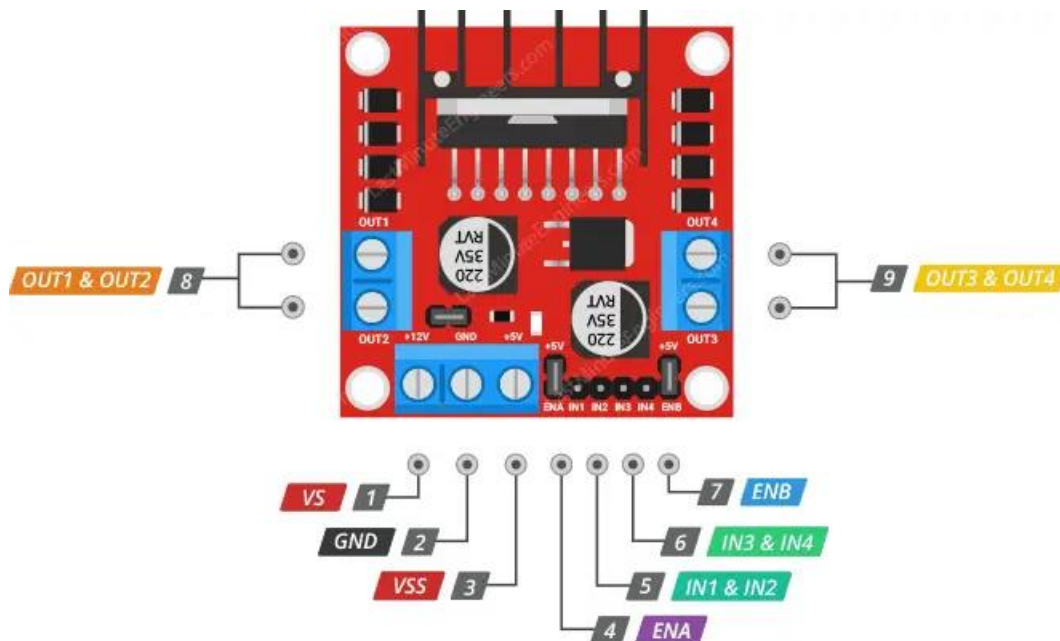
ESP32 MICROCONTROLLER

1. Overview of ESP32
2. Features: Wi-Fi, Bluetooth, Dual-Core Processor
3. Pinout Diagram
4. Programming: Arduino IDE
5. Applications: IoT, robotics, and automation
6. Advantages: Low power consumption, high performance
7. Disadvantages: Steep learning curve
8. ESP32 Board Package
9. Bluetooth Library
10. Example Code



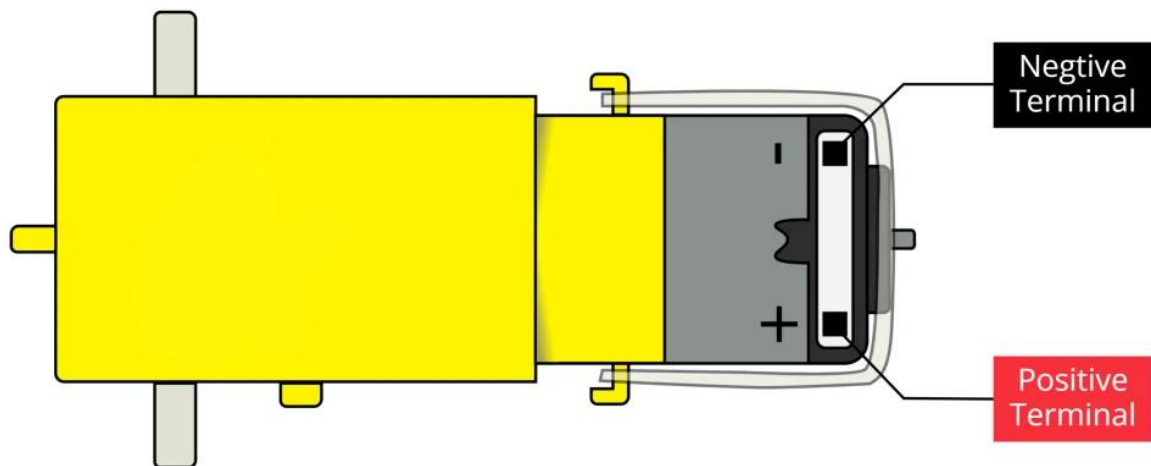
L298N MOTOR DRIVER

1. Overview of L298N
2. Features: Dual H-Bridge, High Current Capacity
3. Pinout Diagram
4. Applications: Robotics, automation, and motor control
5. Advantages: High current capacity, easy to use
6. Disadvantages: Heat generation
7. Connection Diagram
8. Example Code
9. Motor Control: Forward, backward, left, and right
10. Speed Control: PWM



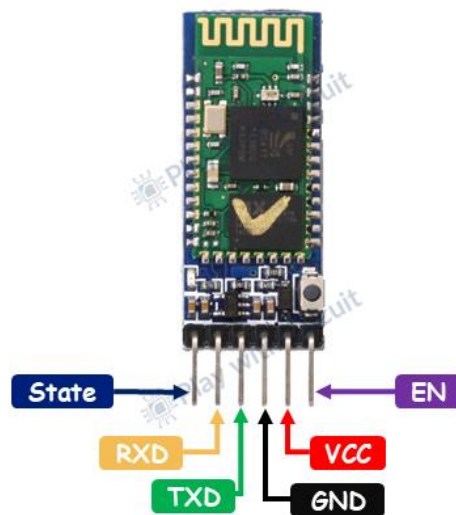
DC MOTORS

1. Overview of DC Motors
2. Features: High Torque, Low Power Consumption
3. Connection Diagram
4. Applications: Robotics, automation, and motor control
5. Advantages: High torque, low power consumption
6. Disadvantages: Limited speed control
7. Motor Control: Forward, backward, left, and right
8. Speed Control: PWM
9. Gear Ratio: 1:10
10. Motor Specifications



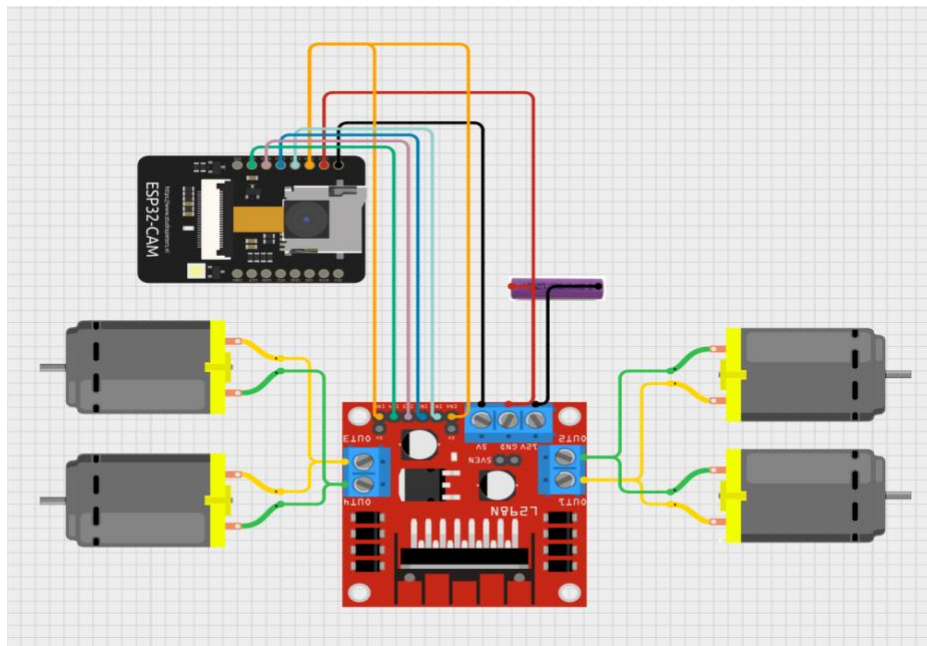
BLUETOOTH MODULE (HC-05)

1. Overview of HC-05(Built in)
2. Features: Bluetooth 2.0, Easy to Use
3. Connection Diagram
4. Applications: Robotics, automation, and IoT projects
5. Advantages: Easy to use, low power consumption
6. Disadvantages: Limited range
7. Pairing: HC-05 with Android device
8. Data Transfer: Serial communication
9. Example Code
10. Bluetooth Library



CIRCUIT CONNECTION

1. Schematic Diagram of the Bluetooth Controlled Car
2. Connection between ESP32, L298N, DC Motors, and Bluetooth Module
3. Power Supply: Battery
4. Jumper Wires
5. Breadboard
6. Car Chassis
7. Wheels and Motor Assembly
8. Car Body
9. Circuit Explanation
10. Troubleshooting

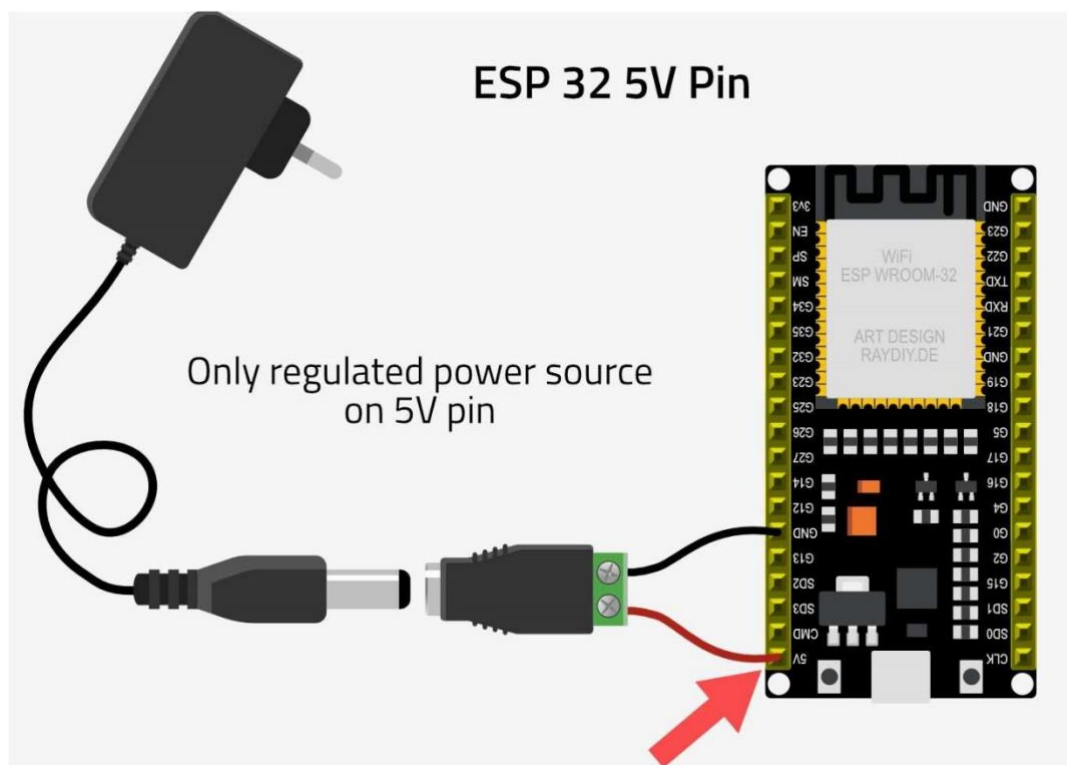


CIRCUIT INTERACTION

- 1. Explanation of the circuit connections**
- 2. How the components interact with each other**
- 3. Power Supply: Battery**
- 4. Jumper Wires**
- 5. Breadboard**
- 6. Car Chassis**
- 7. Wheels and Motor Assembly**
- 8. Car Body**
- 9. Circuit Troubleshooting**
- 10. Debugging**

POWER SUPPLY

1. Overview of the power supply (battery)
2. Connection to the circuit
3. Voltage: 9V
4. Current: 1A
5. Power Consumption: Low
6. Battery Life: 2 hours
7. Charging: USB
8. Power Supply: Battery
9. Jumper Wires
10. Breadboard



SOFTWARE

- 1. Arduino IDE**
- 2. ESP32 Board Package**
- 3. Bluetooth Library**
- 4. Code Explanation**
- 5. Upload Example Code**
- 6. Programming: Arduino IDE**
- 7. Applications: IoT, robotics, and automation**
- 8. Advantages: Easy to use, cross-platform**
- 9. Disadvantages: Limited functionality**
- 10. Alternative: Python, C++**

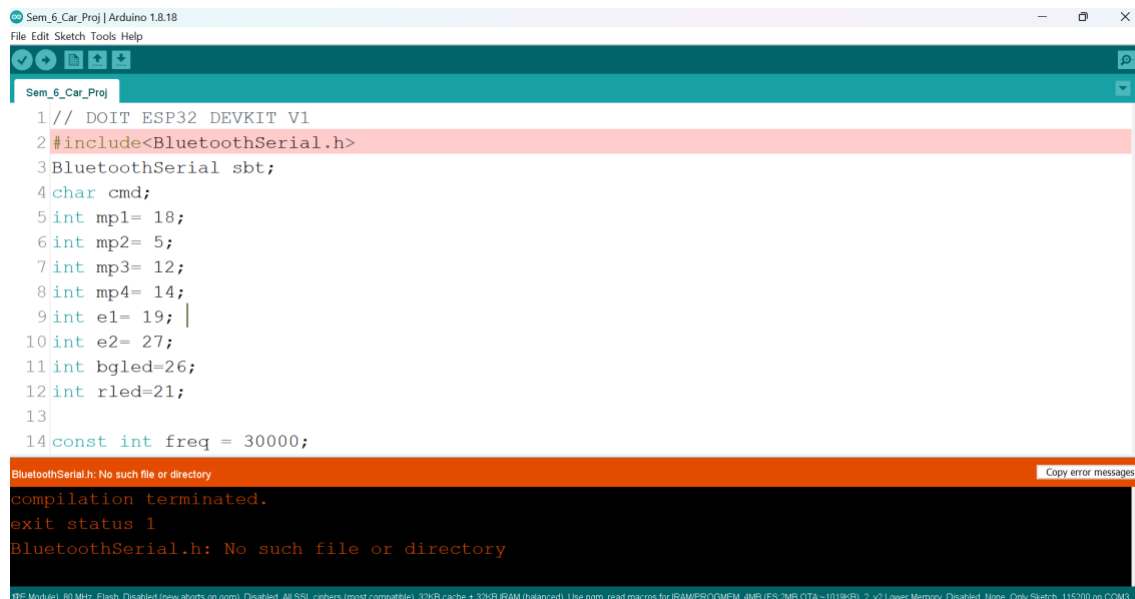


ESPRESSIF



CODE EXPLAINED

1. Explanation of the code used to control the car
2. How the Bluetooth module is used to receive commands
3. Motor control: Forward, backward, left, and right
4. Speed control: PWM
5. Example Code
6. Code Structure: Functions, loops, and variables
7. Variables: Motor speed, direction, and Bluetooth data
8. Functions: Motor control, Bluetooth communication
9. Loops: Main loop, motor control loop
10. Debugging: Serial monitor, Bluetooth debugging



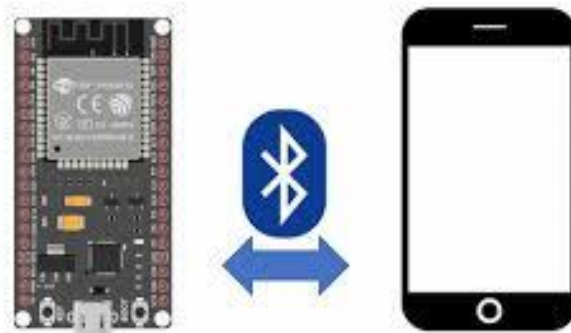
The screenshot shows the Arduino IDE interface. The top bar indicates the project is 'Sem_6_Car_Proj' using 'Arduino 1.8.18'. The code editor displays the following code:

```
1 // DOIT ESP32 DEVKIT V1
2 #include<BluetoothSerial.h>
3 BluetoothSerial sbt;
4 char cmd;
5 int mp1= 18;
6 int mp2= 5;
7 int mp3= 12;
8 int mp4= 14;
9 int e1= 19; |
10 int e2= 27;
11 int bgled=26;
12 int rled=21;
13
14 const int freq = 30000;
```

Below the code editor, an orange error message bar states: 'BluetoothSerial.h: No such file or directory'. The bottom status bar shows: 'BT Module: 80 MHz, Flash, Disabled (new alerts on oom), Disabled, All SSL ciphers (most compatible), 32KB cache + 32KB RAM (balanced), Use pgm_read macros for IRAM/PROGMEM, 4MB (FS 3MB OTA~1019KB), 2 v2 Lower Memory, Disabled, None, Only Sketch, 115200 on COM3'.

BLUETOOTH SETUP

1. Overview of the Bluetooth app used to control the car
2. Features: Button layout, connection establishment
3. Connection: HC-05 with Android device
4. Data Transfer: Serial communication
5. Uploading example Code
6. App: Taken from play store
7. User Interface: Buttons, text views
8. Bluetooth Communication: HC-05, serial communication
9. App Functionality: Motor control, speed control
10. Debugging: GCC compiler, Bluetooth debugging



CAR CHASSIS

- 1. Overview of the car chassis**
- 2. Features: Durable, easy to assemble**
- 3. Materials: Plastic, Paper, Cardboard**
- 4. Dimensions:**
 - Length 8 Inch,**
 - Width 6 Inch,**
 - Height 2 Inch**
- 5. Wheels and Motor Assembly**
- 6. Car Body: Made with cardboard**
- 7. Chassis Design: Simple, cardboard**
- 8. Assembly: Adhesive and Tape**
- 9. Disassembly: Easy, Throw 3-4 times**
- 10. Maintenance: High, check adhesive reguarly**

WHEELS & MOTORS

1. Overview of the wheel and motor assembly
2. Features: High torque, low power consumption
3. Motor Specifications
4. Gear Ratio: 1:10
5. Wheel Specifications
6. Assembly: Easy, screw-based
7. Disassembly: Easy, screw-based
8. Maintenance: Easy, accessible
9. Motor Control: Forward, backward, left, and right
10. Speed Control: PWM



CAR BODY

- 1. Overview of the car body**
- 2. Features: Lightweight, easy to design**
- 3. Materials: Plastic, Paper, Cardboard**
- 4. Dimensions:**
 - Length 8 Inch,**
 - Width 6 Inch,**
 - Height 2Inch**
- 5. Design: Simple**
- 6. Assembly: Easy, Adhesive and Tape**
- 7. Disassembly: Easy, Throw 3-4 times**
- 8. Maintenance: Easy, accessible**
- 9. Aesthetics:**
 - Color- White,**
 - Shape- Rectangle,**
 - Design- Good**
- 10. Functionality: Motor control**

ASSEMBLY & TESTING

- 1. Step-by-step assembly process**
- 2. Testing and debugging**
- 3. Motor control: Forward, backward, left, and right**
- 4. Speed control: PWM signals**
- 5. Bluetooth communication: HC-05, serial communication**
- 6. App functionality: Motor control, speed control**
- 7. Troubleshooting: Serial monitor, BLE debugging**
- 8. Debugging: GCC compiler, Bluetooth debugging**
- 9. Testing: Motor control, speed control**

RESULTS

- 1. Demonstration of the Bluetooth-controlled car**
- 2. Results of the project**
- 3. Motor control: Forward, backward, left, and right**
- 4. Speed control: PWM**
- 5. Bluetooth communication: HC-05, serial communication**
- 6. App functionality: Motor control, speed control**
- 7. Troubleshooting: Serial monitor, Bluetooth debugging**
- 8. Debugging: GCC compiler , Bluetooth debugging**
- 9. Testing: Motor control, speed control**
- 10. Validation: Working Successfully**

ADVANTAGES

- 1. Advantages of the Bluetooth-controlled car**
- 2. Upcoming Potential applications**
- 3. Robotics and automation usage**
- 4. Can be used in IoT projects**
- 5. Educational and learning purposes**
- 6. Easy to use and control**
- 7. Low power consumption**
- 8. High peak torque**
- 9. Simple design**
- 10. Affordable(Used 2200 rupees)**

DISADVANTAGES

- 1. Limitations of the project**
- 2. Future scope not clear**
- 3. Limited range due to bluetooth**
- 4. Interference from other devices can be possible**
- 5. Limited speed control**
- 6. Limited motor control**
- 7. Dependence only on Bluetooth communication**
- 8. Limited app functionality**
- 9. Limited debugging capabilities**
- 10. Limited scalability**

CONCLUSION

- 1. Summary of the project**
- 2. Bluetooth-controlled car using ESP32 and L298N**
- 3. Motor control: Forward, backward, left, and right**
- 4. Speed control: PWM**
- 5. Bluetooth communication: HC-05, serial communication**
- 6. App functionality: Motor control,**

THANKS

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